(Very) Quick Look Results

- Three HST image triples (619, 631, 645 nm) for cloud pressure and ammonia abundance are 'analyzed'
- Times (2025-10-16) and notable features:
  - ~0015Z – GRS, NEDFs, belts/zones
  - ~0325Z – SED NEDFs
  - ~0500Z – NEDFs
- Qualitative analysis and comparison to ground data
  - Ratio of CH_4 transmission *over* NH_3 transmission gives *qualitative* indication NH_3 abundance
  - Co-alignment of "reg" maps done by hand
  - COTS software (QFitsView, MaximDL) used

Considerations

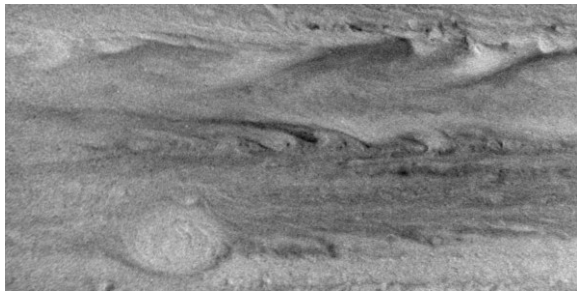
- Simple ratios of transmission computed
  - $T_{CH4} = R_{619}/R_{631}$, $T_{NH3} = R_{645}/R_{631}$
  - Cloud Pressure ~ $1-T_{CH4}$
  - Ammonia ~ T_{CH4}/T_{NH3}
- Possible navigation issues with ground data for Sys 3 long
- No accounting for contamination of 645 NH_3 band by 668 CH_4 band
- Sensitive to pressures of ~2 bar

GRS, SEB, and NEDFs – HST vs Ground

- NEB is NH_3 depleted
- **NH_3 enhancements extend northward 'over' NEDFs**
- GRS shows high clouds and low NH_3
- SEB shows deep clouds with NH_3 depletion
- STrZ shows higher clouds *and* enhanced NH_3

Ammonia ~ T_{CH4}/T_{NH3}

Cloud Press. ~ $1-T_{CH4}$



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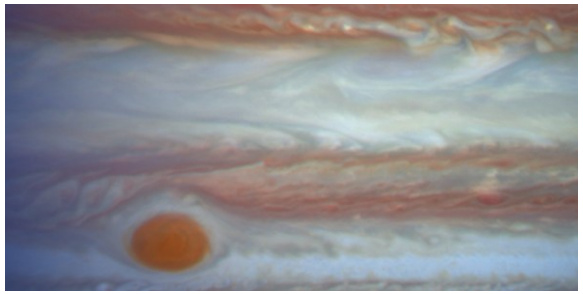

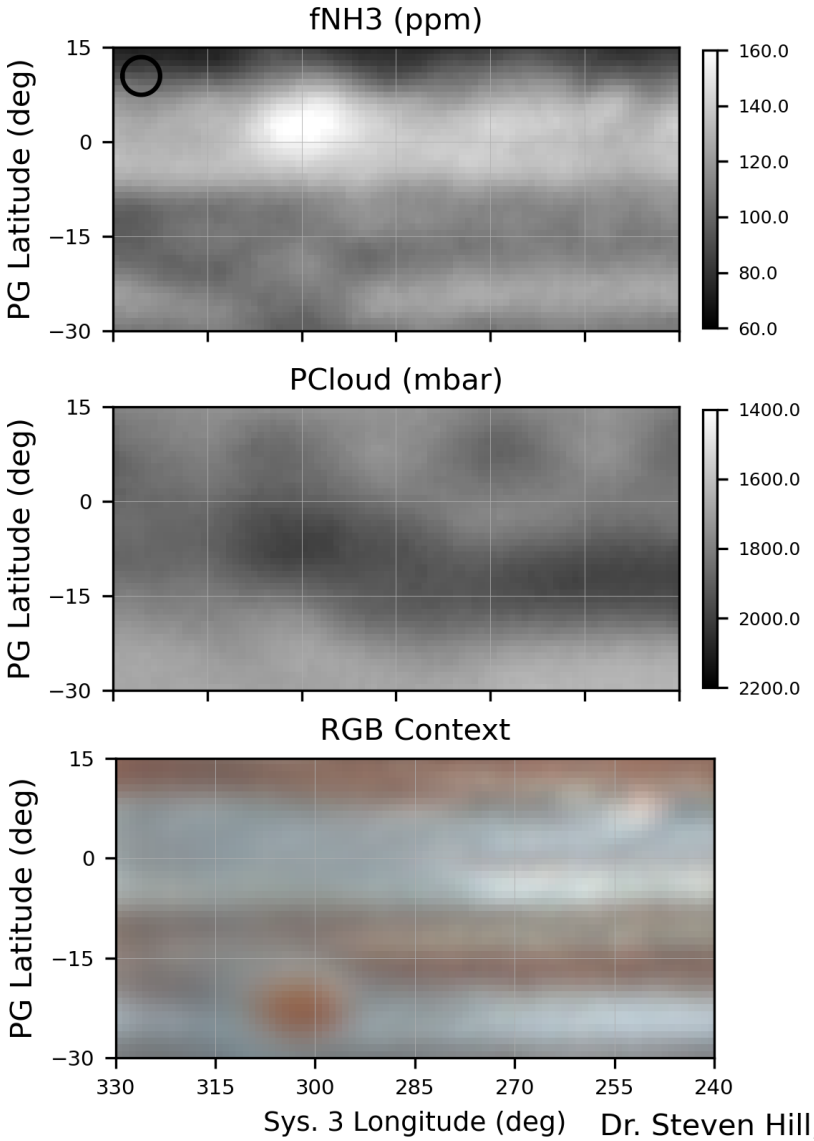

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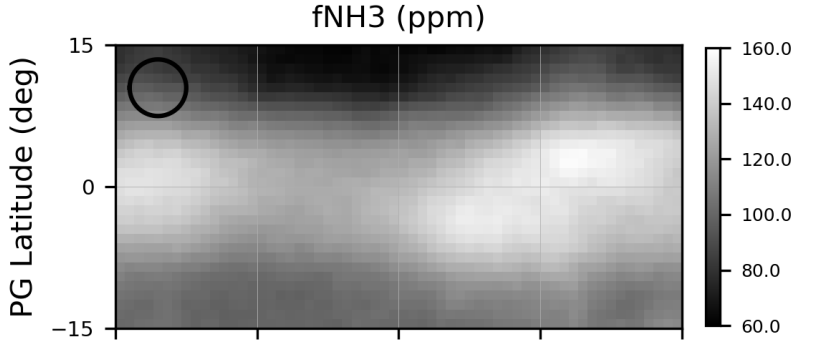
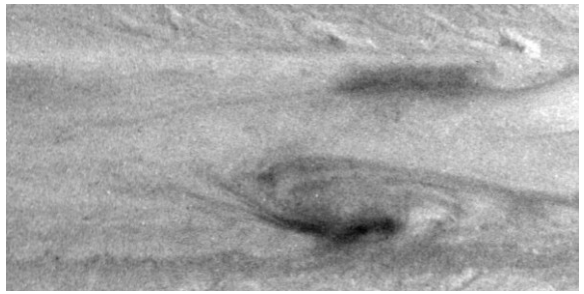
SED and NEDFs – HST vs Ground

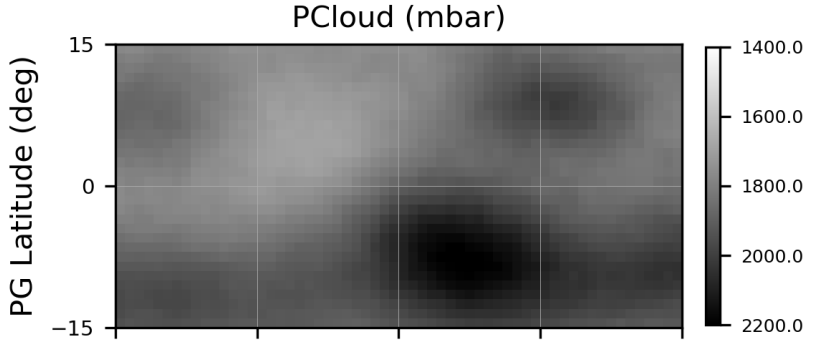
- NEB is NH_3 depleted
- **NH_3 enhancements extend northward 'over' NEDFs**
- SED shows *very* deep clouds, equatorward NH_3 enhancement
- SED also shows a unique greenish tinge in context image

Ammonia ~ T_{CH4}/T_{NH3}



Cloud Press. ~ 1-T_{CH4}





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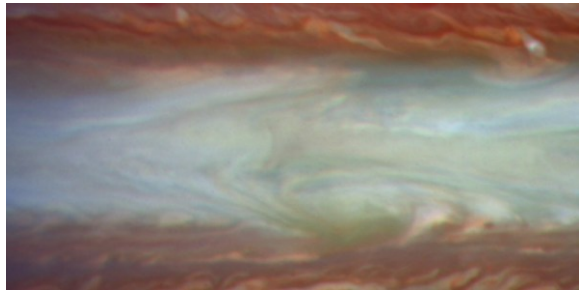

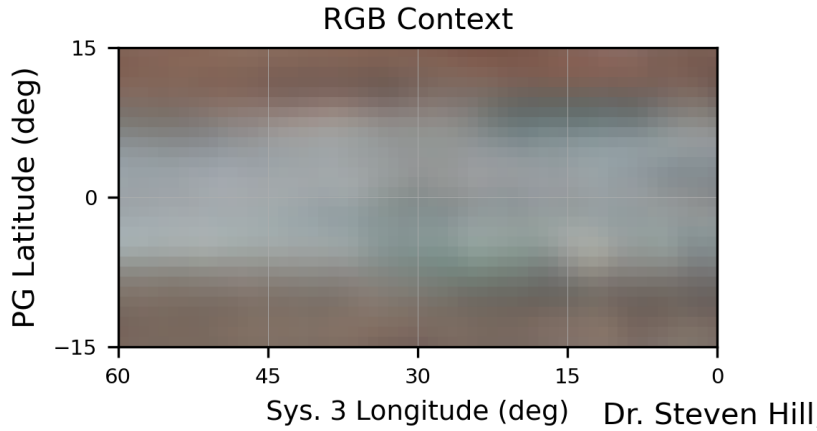

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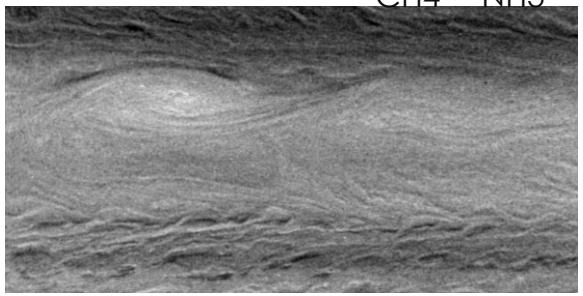
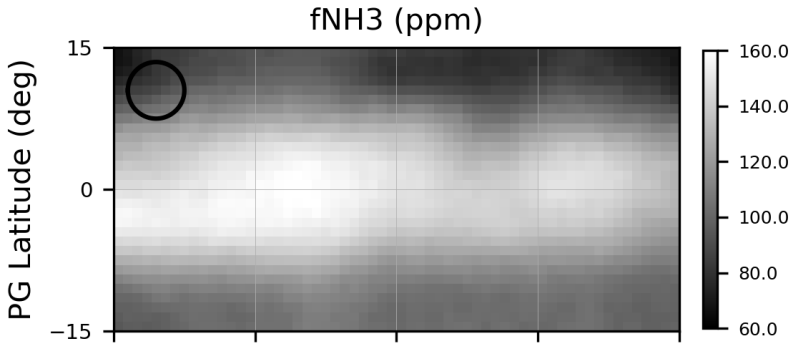
NEDFs – HST vs Ground

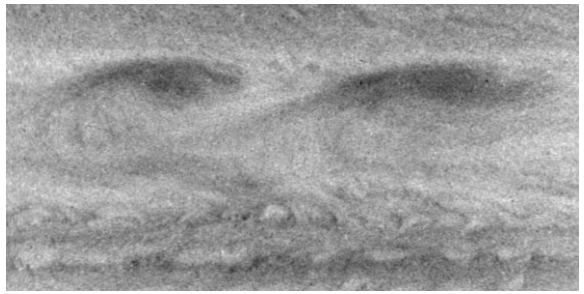
- NEB is NH_3 depleted
- **NH_3 enhancements extend northward 'over' NEDFs**
- SEB shows very deep clouds and some NH_3 depletion
- STrZ shows higher clouds *and* enhanced NH_3

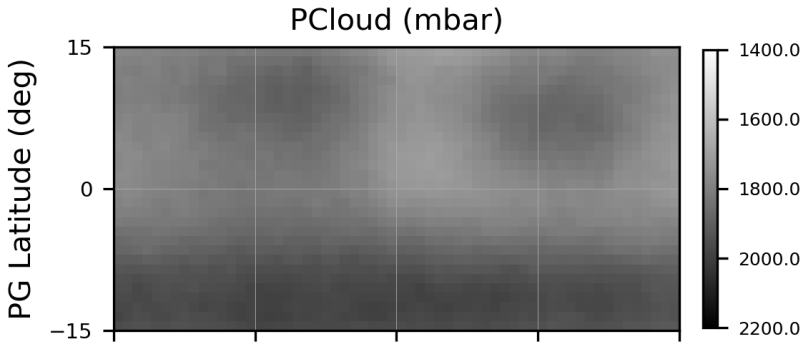
Ammonia ~ T_{CH4}/T_{NH3}





Cloud Press. ~ 1-T_{CH4}





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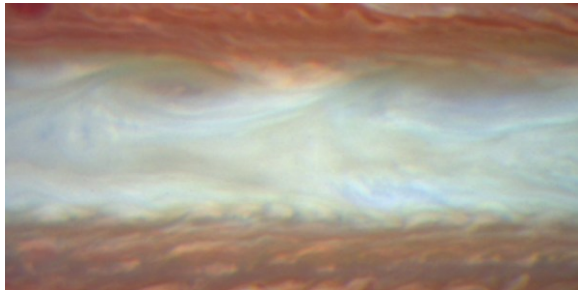

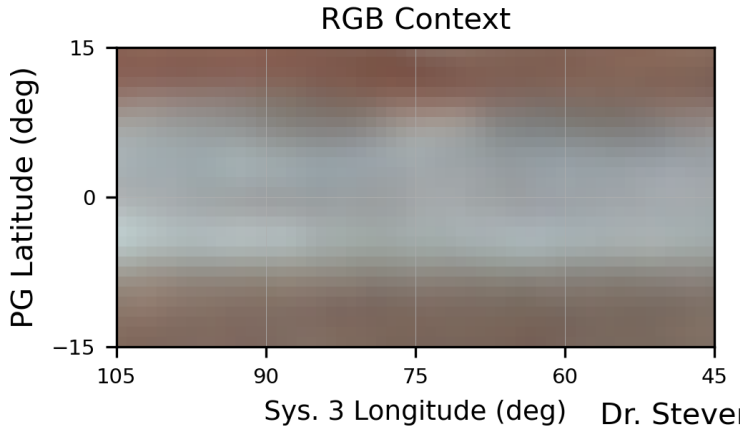

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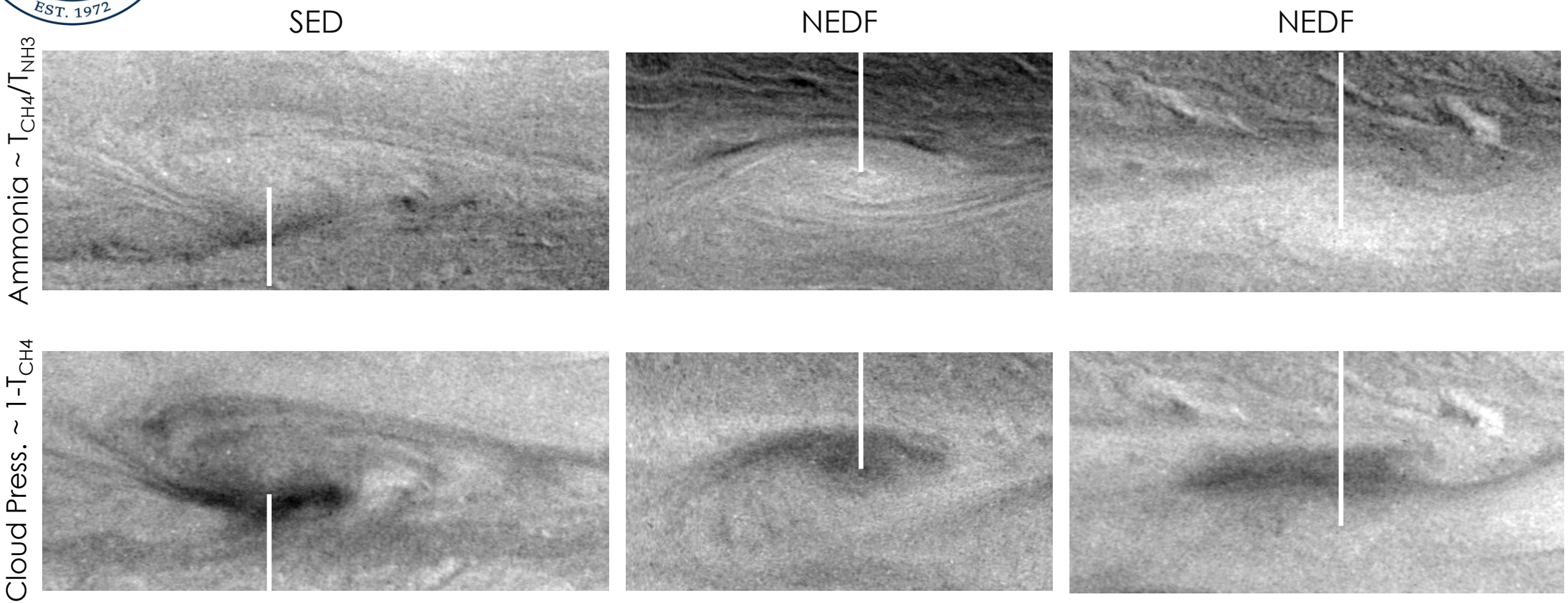
HST Close Ups of SED and NEDFs



- NH_3 enhancement is equatorward of NEDFs , but overlaps NEDFs
- Is the SED a wave number 1 Rossby Wave Feature?

Thoughts

- NH_3 enhancements have centroids south of NEDFs seen in ground and HST data
- HST data show the enhancements are spatially extended and that the extended nature seen in ground data is not due to poorer resolution
- While the enhancements are extended, structured features with centers are south of the NEDFs, they also extend poleward and overlap the NEDFs
- The SED looks like a north-south mirror version of NEDFs providing further support that the SED is the result of a Rossby wave with wave number = 1